

Senne Vleminckx, PhD

Health data science · nursing workforce systems · epidemiological modelling

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PROFILE

Postdoctoral researcher in health data science, combining a decade of clinical practice in emergency and intensive care nursing with quantitative methodology. My doctoral work developed the Balanced Care Teams framework and the Structured Collaboration Index, a roster-derived measure of collaboration structure in multi-team systems, and linked team composition to patient and staff outcomes across a four-country cohort. I now extend this work towards infectious-disease transmission modelling, agent-based simulation and health-economic evaluation, and work as an independent statistical consultant for healthcare organisations and research groups.

RESEARCH LINES

- **Nursing workforce systems.** Team composition, heterogeneity, stability and collaboration structure, and their association with patient safety and staff sustainability (BCT framework; Structured Collaboration Index).
- **Measurement and psychometrics.** Development and validation of instruments for workforce and care-team constructs, using EFA/CFA, IRT, and polychoric-WLSMV estimation.
- **Simulation and health-economic modelling.** Agent-based and compartmental transmission models, and cost-effectiveness analysis under partial identification.

EDUCATION

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| 2026 | PhD in Medical Sciences , University of Antwerp
Thesis: <i>Balanced Care Teams: A Systems Approach to Nursing Workforce Design. From Conceptual Framework to Empirical Validation and Practice Translation</i> . Public defence 2 July 2026.
Supervisors: Prof. P. Van Bogaert, Prof. K. De Meulenaere, Prof. L. Willem, Prof. F. Haegdorens.
doi:10.63028/10067/2241000151162165141 |
| 2019 | Master in Nursing & Midwifery , University of Antwerp |
| 2015 | Bachelor in Intensive and Emergency Care , Hogeschool UCLL |
| 2014 | Bachelor in Nursing , Hogeschool UCLL |

APPOINTMENTS

Academic

Postdoctoral Researcher

University of Antwerp; from Oct 2026

Family Medicine and Population Health (FAMPOP), ACCELERATE project, with Prof. L. Willem

- Agent-based and compartmental transmission modelling, with health-economic evaluation of intervention strategies.

PhD Fellow

University of Antwerp; Oct 2022 – Sep 2026

Balanced Care Teams: A Systems Approach to Nursing Workforce Design

- Developed a holistic framework describing nursing team composition through capacity, processes and demands, extending ratio-based staffing models with heterogeneity, stability and structured collaboration.
- Constructed the Structured Collaboration Index, a roster-derived metric quantifying the stability of individual interaction patterns within meta-teams and sub-teams; developed on 36 months of roster data across 30 wards.
- Led a four-country observational study (48 wards, approximately 1,200 nurses) linking team composition to patient safety and staff sustainability, using multilevel models with multiple imputation across 10 imputed datasets.

- Established prevalence and drivers of multiple job holding among Belgian nurses (38.3% prevalence; 64.2% push-driven), informing national workforce policy debate.

Junior Researcher (Project Lead)

University of Antwerp; Oct 2019 – Oct 2022

Balanced Nursing Teams (BNuT) project: decision-support platform integrating administrative hospital data with 360-degree survey data

- Designed and validated a 360-degree nursing team survey, deployed across more than 100 wards in 18 healthcare organisations.
- Co-developed the back-end data architecture for the BNuT platform.
- Led procurement and contract negotiations with an external software development firm, and coordinated the academic-industry collaboration.
- Managed integration of multi-hospital administrative data, including coordination with a data-integration firm and compliance with GDPR and ethics requirements.
- Presented the project to venture capitalists and institutional stakeholders to explore funding routes.

Clinical

Emergency Nurse

Ziekenhuis Netwerk Antwerpen (now ZAS); Oct 2019 – May 2022

Urban emergency department, >110 patients/day. COVID-19 pandemic preparedness; prehospital care.

Intensive Care Nurse

University Hospital Leuven; Mar 2016 – Sep 2019

Tertiary medical ICU. Led an improvement project (master's thesis) that reduced catheter-related infection incidence by 80%.

Emergency Nurse

University Hospital Leuven; Jul 2015 – Feb 2016

Tertiary emergency department.

PUBLICATIONS

Peer-reviewed articles

Vleminckx, S., De Meulenaere, K., Van Bogaert, P., Abrams, S., Willem, L., and Haegdorens, F. (Sept. 2026). "Beyond staffing ratios – team demographic diversity, workforce stability, and their effects on patient safety and workforce sustainability: a multi-country analysis". In: *International Journal of Nursing Studies* 181, p. 105570. ISSN: 0020-7489. DOI: [10.1016/j.ijnurstu.2026.105570](https://doi.org/10.1016/j.ijnurstu.2026.105570).

Vleminckx, S., Van Bogaert, P., De Keyser, W., and Haegdorens, F. (Aug. 2026). "From concept to practice: lessons from the Balanced Nursing Teams decision-support system". In: *JMIR Nursing* 9, e92417. ISSN: 2562-7600. DOI: [10.2196/92417](https://doi.org/10.2196/92417).

Van de Weyer, G., Haegdorens, F., **Vleminckx, S.**, Hilderson, D., and Franck, E. (May 2026). "Self-confidence and anxiety in clinical decision-making among nursing students during high-fidelity simulation training: a longitudinal study". In: *Advances in Simulation* 11, p. 50. DOI: [10.1186/s41077-026-00445-8](https://doi.org/10.1186/s41077-026-00445-8).

Vleminckx, S., Sevenans, A., Bouchatta, M., Verbeeck, I., Franck, E., and Haegdorens, F. (May 2025). "Multiple job holding and its influencing factors among Belgian nurses: a cross-sectional study". In: *Health Policy* 155, p. 105288. DOI: [10.1016/j.healthpol.2025.105288](https://doi.org/10.1016/j.healthpol.2025.105288).

Haegdorens, F., Sloomans, S., **Vleminckx, S.**, Van Bogaert, P., and Franck, E. (Mar. 2025). "Turnover and intention to leave among Belgian healthcare workers after COVID-19: a two-year longitudinal cohort study". In: *Journal of Advanced Nursing*. DOI: [10.1111/jan.16925](https://doi.org/10.1111/jan.16925).

Vleminckx, S., Van Bogaert, P., De Meulenaere, K., Willem, L., and Haegdorens, F. (Sept. 2024). “Factors influencing the formation of balanced care teams: the organisation, performance, and perception of nursing care teams and the link with patient outcomes: a systematic scoping review”. In: *BMC Health Services Research* 24.1, p. 1129. DOI: [10.1186/s12913-024-11625-5](https://doi.org/10.1186/s12913-024-11625-5).

Vleminckx, S., Van Bogaert, P., Daneels, K., Proost, A., Sarens, S., and Haegdorens, F. (May 2024). “Strategies and interventions to support quality outcomes in the home care setting: a longitudinal multilevel study”. In: *Journal of Nursing Administration* 54.5, pp. 278–285. DOI: [10.1097/NNA.0000000000001424](https://doi.org/10.1097/NNA.0000000000001424).

Manuscripts under review

Vleminckx, S., De Meulenaere, K., Haegdorens, F., Van Bogaert, P., and Willem, L. (2025). “The Structured Collaboration Index: a methodological approach for measuring collaboration intensity in multi-team systems”. In: *Small Group Research*. Submitted.

Averens, T., **Vleminckx, S.**, Haegdorens, F., and Franck, E. (2026). “Development and Psychometric Evaluation of a Multidimensional, Multilevel Scale of Nurse Autonomy in Acute Hospital Care”. In: *Journal of Advanced Nursing*. Submitted.

Janssens, N., Van den Berghe, S., Goossens, E., Vanden Wyngaert, K., **Vleminckx, S.**, Vyverman, L., Van Wilder, L., and De Smedt, D. (2026). “Psychometric validation of the Flemish version of the Transition Readiness Assessment Questionnaire (TRAQ 6.0) in young people with chronic conditions”. Submitted.

Antonino, L., Goossens, E., Olmen, J. van, Verhulst, S., **Vleminckx, S.**, Van Hoorenbeeck, K., and Antwerp Pediatric Asthma Network Consortium (2025). “Reducing delays in systemic corticosteroid administration in children with asthma: impact of implementing a multi-centre evidence-based clinical pathway. A quasi-experimental study”. Submitted.

Slootmans, S., Van Bogaert, P., **Vleminckx, S.**, and Franck, E. (2025). “Development and validation of an instrument to assess nursing team contexts and competencies regarding continuous quality improvement culture”. Submitted.

Manuscripts in preparation

Arntz, A., **Vleminckx, S.**, Mareen, H., and Haegdorens, F. (2025). “The effect of lighting conditions on video-assessed clinical deterioration compared to live observations using NIPDS: a pilot study”. In preparation.

Datasets & other research outputs

Vleminckx, S., De Meulenaere, K., Van Bogaert, P., Abrams, S., Willem, L., and Haegdorens, F. (Sept. 2026). *Beyond staffing ratios: a multi-country observational study linking nursing team composition with patient and staff outcomes*. Version 2. Dataset. DOI: [10.17632/bbv9bb7rgw.2](https://doi.org/10.17632/bbv9bb7rgw.2).

Massaer, M., Van Damme, R., Franck, E., Timmermans, O., and **Vleminckx, S.** (2026). *Quantifying the risk of perinatal depression associated with a history of depressive episodes: a systematic review and meta-analysis using prediction intervals*. PROSPERO registration CRD420261323894. Systematic review protocol. URL: <https://www.crd.york.ac.uk/PROSPERO/view/CRD420261323894>.

SOFTWARE & SELECTED PROJECTS

BNuT: Balanced Nursing Teams platform

app.bnut.eu

Web application centralising and visualising nursing data across healthcare institutions; integrates survey and administrative data into interactive dashboards for evidence-based staffing decisions. Role: co-design of back-end data architecture, survey instrument, and analytics layer.

Structured Collaboration Index: reference implementation

Apache-2.0

Python implementation of the SCI for deriving collaboration-intensity metrics from nursing rosters, with worked examples and tests.

RAG-based student query system

internal

Retrieval-augmented generation system answering student queries from internal university documents (regulations, procedures), reducing routine administrative load.

Citation verification tool

internal

Automated academic-integrity tool validating citations in student papers against source databases, supporting grading and reference accuracy.

ABM dashboard prototype

internal

Interactive dashboard for real-time visualisation of agent-based model output and policy-scenario comparison, supporting methodological development in healthcare policy optimisation.

TEACHING & SUPERVISION

Teaching

- 2022 – present **Advanced Statistics**, MSc Nursing & Midwifery, University of Antwerp. Teaching assistant on an elective course (4 ECTS, 18 contact hours shared between two teachers). Topics: multilevel regression, latent variable modelling, Markov models, MCMC, machine learning.
- 2025 – present **Statistics tutor**, University of Antwerp. Individual and small-group support for master's students on study design, analysis and interpretation.
- 2024 – present **Artificial Intelligence in Healthcare**, guest lecture in the MSc course on Academic Development, University of Antwerp. 4 contact hours; cohort of 200–300 students.

Doctoral supervision

- 2025 – present **E. Claes**, daily supervisor (*begeleider*): “Optimalisatie van spoedgevallenzorg via datagedreven beslissingen” [Optimising emergency care through data-driven decision-making]

Master's thesis supervision

MSc Nursing and Midwifery, University of Antwerp

- 2025 – 2026 Arntz, A.: “The effect of lighting conditions on video-assessed clinical deterioration compared to live observations using NIPDS: a pilot study”
- 2025 – 2026 Dijckmans, E.: “Ontwikkeling en validatie van een kennistest voor competenties omtrent artificiële intelligentie in de gezondheidszorg bij artsen en studenten geneeskunde” [AI knowledge test for physicians and medical students]
- 2025 – 2026 Installé, S.: “Datagedreven leiderschap in Vlaamse ziekenhuizen, zijn leidinggevendenden mee?” [Data-driven leadership in Flemish hospitals]
- 2025 – 2026 Jansen, J.: “Ontwikkeling en validatie van een kennistest voor competenties in basisprincipes van artificiële intelligentie bij verpleegkundigen en studenten verpleegkunde” [AI knowledge test for nurses and nursing students]
- 2024 – 2025 De Schrijver, C.: “Ontwikkeling van een meetinstrument voor datagedreven besluitvorming bij (adjunct-)leidinggevendenden uit de Vlaamse gezondheidszorg: een gemodificeerde Delphi-studie met pilootonderzoek” [Instrument development for data-driven decision-making among healthcare managers]
- 2024 – 2025 Van Oorschoot, C.: “Factoren geassocieerd met absentisme bij zorgpersoneel in een Belgisch ziekenhuisnetwerk: een cross-sectioneel onderzoek” [Factors associated with absenteeism among healthcare staff]
- 2023 – 2024 Neyens, L.: “Decision support tijdens triage op spoedgevallen: ontwikkeling van een machine learning model voor patiëntenallocatie en interventies” [Machine learning decision support for emergency triage]
- 2022 – 2023 Michiels, A.: “De ligduur van patiënten op spoedgevallen: een verkennende studie naar voorspellende factoren” [Predictors of emergency department length of stay]

MSc Organisation and Management, University of Antwerp

- 2024 – 2025 Goos, M. & Vanuffelen, Z.: “Opvolging pre-implementatieproces van BNuT binnen AZ Klina” [Monitoring the BNuT pre-implementation process]

MSc Applied Economic Sciences: Business Administration, University of Antwerp

- 2021 – 2022 Van Reusel, C.: “Software spin-off: opstelling marketing plan en business model canvas” [Software spin-off: marketing plan and business model canvas]

CONSULTANCY & ADVISORY

- 2026 – present **Independent statistical consultant**, registered self-employed, Belgium. Study design, sample size and power, multilevel and Bayesian modelling, psychometric validation, and reproducible analysis pipelines for healthcare organisations and research groups.
- 2024 – present **Statistical consultant**, Centre for Research and Innovation in Care (CRIC), University of Antwerp. Methodological support on design, analysis and reporting.
- 2025 **External expert**, “Pilot study on a new payment system for home nursing” (Van den Heede et al.).

PEER REVIEW

Reviewer (2024–2026) for: BMC Medical Education; BMC Nursing; International Journal of Nursing Studies Advances; International Nursing Review; Journal of Advanced Nursing; Journal of Happiness Studies; Nursing Outlook.

Verified through [ORCID](#) and Web of Science Reviewer Recognition.

INVITED TALKS & CONFERENCE CONTRIBUTIONS

Invited talks

- 2025 *Artificial intelligence applications in nursing research*. Keynote, Belgische Vereniging voor Verpleegkundige Wetenschap (BVVS) online symposium.
- 2025 *Development of a retrieval-augmented generation system for handling student questions*. CRIC Symposium, Antwerp.
- 2024 *Development of a novel metric to quantify collaboration intensity in multi-team systems*. CRIC Symposium, Antwerp.

Conference contributions

- 2025 *Understanding the influencing factors for multiple job holding in nursing*. International Council of Nurses (ICN) Congress, Helsinki. Poster.
- 2024 *Development of a novel metric to quantify collaboration intensity in multi-team systems*. European Institute for Advanced Studies in Management (EIASM) Congress, Barcelona. Oral.
- 2024 *Factors influencing the formation of balanced care teams*. Care4 Congress, Antwerp. Poster.

METHODOLOGICAL & TECHNICAL SKILLS

Design & inference	Multilevel/hierarchical modelling, Bayesian inference (MCMC/HMC, prior specification, posterior predictive checking), regularised regression, survival analysis, missing data and multiple imputation, uncertainty quantification, partial identification
Measurement	EFA/CFA, structural equation modelling, IRT and Rasch analysis, polychoric correlations with WLSMV estimation, reliability (ω , ECV), instrument development and cross-cultural validation
Modelling	Compartmental transmission models (SIR/SEIR), stochastic formulations (CTMC, Gillespie), agent-based modelling, health-economic evaluation (CEA/CUA, QALY/DALY, ICER/NMB, PSA, value of information)
Machine learning	Tree ensembles and random forests, model validation and calibration, Bayesian optimisation
Programming	Python (pandas, NumPy, SciPy, scikit-learn, PyMC/NumPyro, ArviZ, BoTorch, Trieste, Mesa, Streamlit), SQL/PostgreSQL, R, SPSS, $\LaTeX$, Quarto, Git/GitHub
Infrastructure	High-performance computing (VSC cluster), reproducible workflows, data engineering and pipeline design
Research tools	Qualtrics (survey design), RevMan (systematic reviews), Zotero/BibTeX
Domain expertise	Nursing workforce analytics, team composition metrics, emergency and intensive care practice, healthcare data governance and GDPR compliance

LANGUAGES

Dutch	Native
English	Fluent; working language for research, teaching and publication

RESEARCH STAYS & ADDITIONAL TRAINING

2025 – 2026	Visiting researcher , Erasmus MC DataHub, Rotterdam. Host: Prof. dr. M. van Genderen. Machine learning for ICU patient acuity assessment and resource allocation.
2026	ESSA Summer School: Introduction to Agent-Based Modelling, The James Hutton Institute, Aberdeen
2025	Summer School on Bayesian Optimisation, UHasselt
2025	Summer School on Epidemiological Modelling & Health Economics, University of Antwerp
2024	Introduction to AI and Machine Learning for Biomedical Research, VAIA
2023	Micro Organisational Behaviour, KU Leuven
2022	Leadership in Nursing Excellence summer school, University of Antwerp
2022	Introduction to High-Performance Computing, University of Antwerp
2021	Deep Dive into Business, Antwerp Management School